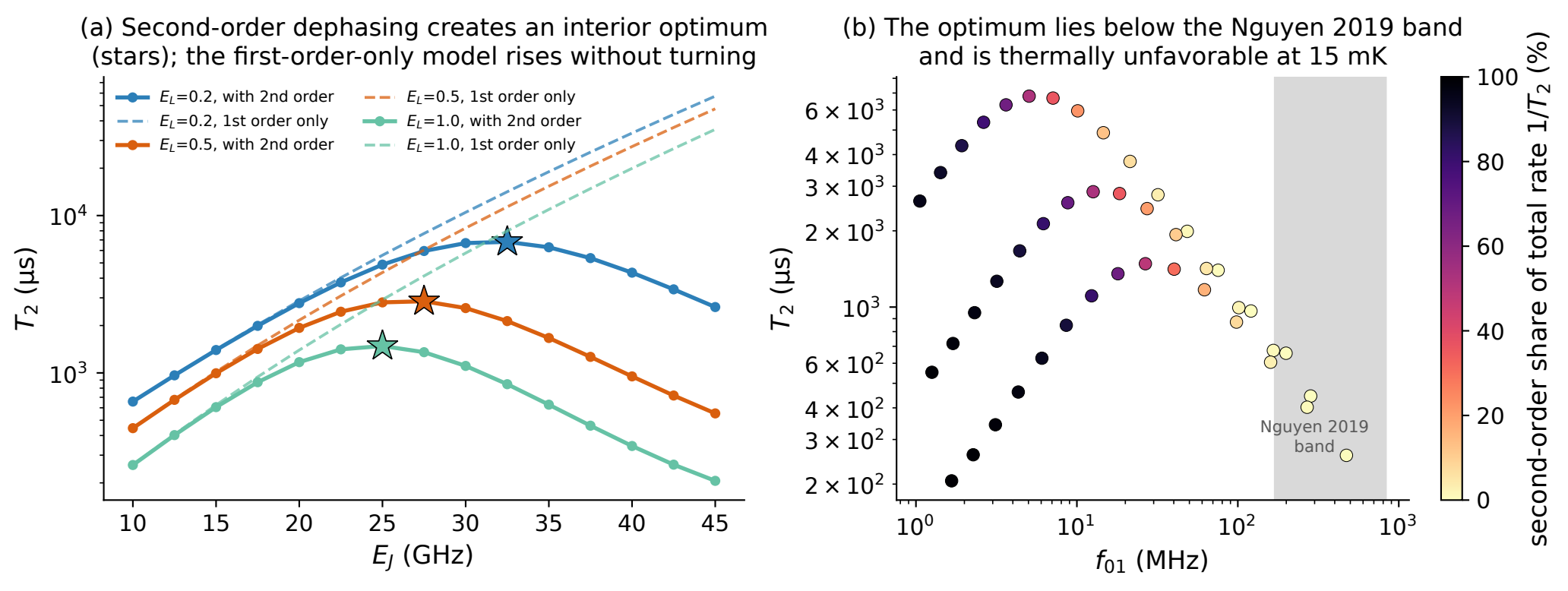

(a) Second-order dephasing creates an interior optimum (stars); the first-order-only model rises without turning
E_L=0.2, with 2nd order
E_L=0.2, 1st order only
E_L=0.5, with 2nd order
E_L=0.5, 1st order only
E_L=1.0, with 2nd order
E_L=1.0, 1st order only
T_2 (μs)
10^4
10^3
10
15
20
25
30
35
40
45
E_J (GHz)
(b) The optimum lies below the Nguyen 2019 band and is thermally unfavorable at 15 mK
T_2 (μs)
6×10^3
4×10^3
3×10^3
2×10^3
10^3
6×10^2
4×10^2
3×10^2
2×10^2
Nguyen 2019 band
10^0
10^1
10^2
10^3
f_{01} (MHz)
second-order share of total rate $1/T_2$ (%)
100
80
60
40
20
0